

MAGNETIC RESONANCE IMAGING

Lecture note



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MRI



INTRODUCTION TO MRI

- **What is MRI?**

- A non-invasive imaging technique used to visualize detailed internal structures of the body.
- Uses strong magnetic fields and radio waves to generate images.

- **Key Advantages:**

- No ionizing radiation (unlike X-rays or CT scans).
- High-resolution images of soft tissues (e.g., brain, muscles, ligaments).

- **Common Uses:**

- Diagnosing tumors, strokes, joint injuries, and neurological disorders.



BASIC PRINCIPLES OF MRI

- **Nuclear Magnetic Resonance (NMR):**
 - MRI is based on the principles of NMR.
 - Protons in hydrogen atoms (abundant in water and fat) align with a strong magnetic field.
- **Radiofrequency (RF) Pulses:**
 - RF pulses disrupt the alignment, causing protons to emit signals.
- **Signal Detection:**
 - Detectors capture the emitted signals, which are processed to create images.



MAGNETIC RESONANCE IMAGING

- The MRI introduced in the early 1980s, is the most recent addition to medical imaging techniques.
- This technique utilizes the magnetic properties of the nucleus to provide images of internal body organs with detailed information about soft-tissue structure.
- The imaging techniques is more complicated comparing to the CT scan imaging.



NUCLEAR MAGNETIC RESONANCE

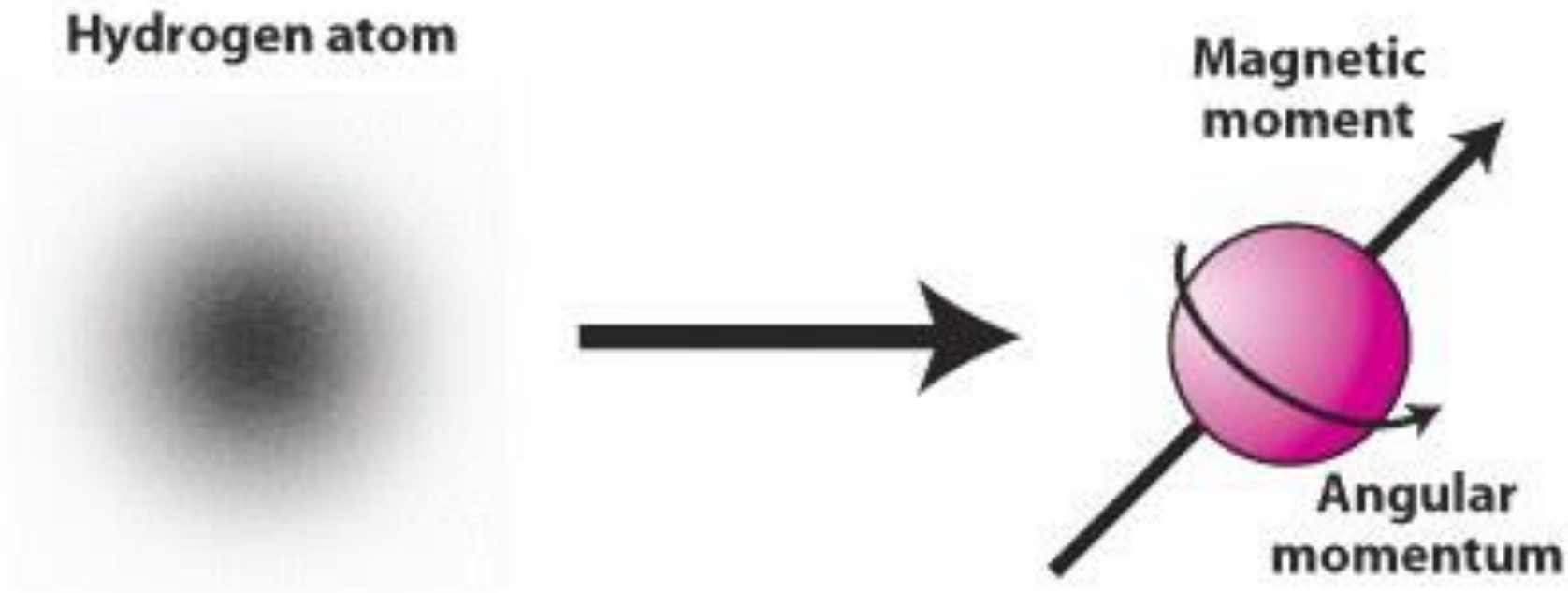
- Protons and neutrons which are the constituents of atomic nuclei possess the quantum mechanical property of spin which has magnitude and direction.
- Imagine these particles as if they were small spinning tops. As a result of spin, the nuclear particles act as small bar magnets.



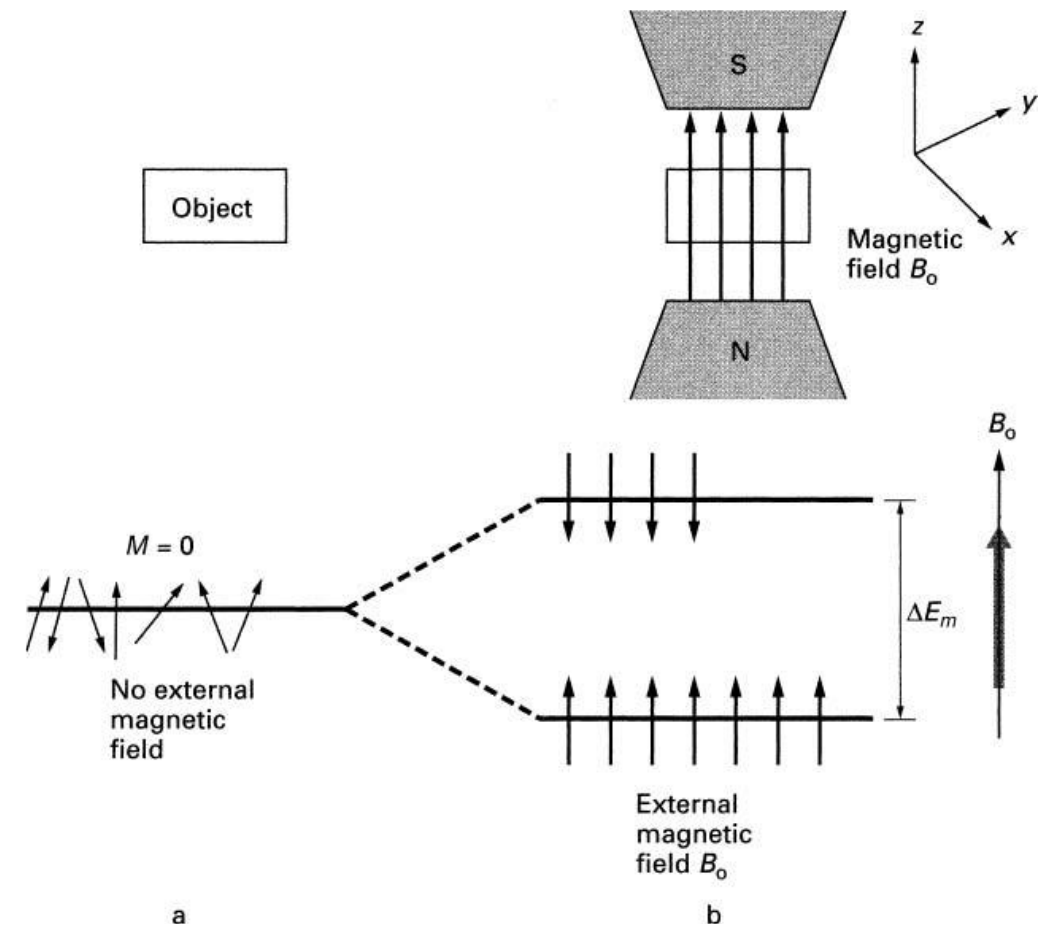
- Inside the nucleus, these small magnets associated with the nucleons (protons and neutrons) line up so as to cancel each other's magnetic fields.
- If the number of nucleons is odd, the cancellation is not complete, and the nucleus possesses a net magnetic moment.
- Nuclei with an odd number of nucleons behave as tiny magnets.



- Hydrogen, which has a nucleus consisting of a single **proton**, does, of course, have a nuclear magnetic moment.



- Normally, the little nuclear magnets in bulk material are randomized in space as shown in part a.
- When an external magnetic field is applied to a material possessing nuclear magnetic moments, the tiny nuclear magnets line up either parallel or antiparallel with the magnetic field as shown in part b



MRI MACHINE COMPONENTS

- **Main Components:**
 - **Magnet:** Generates a strong, uniform magnetic field (0.5–3 Tesla for clinical use).
 - **Gradient Coils:** Create variations in the magnetic field for spatial encoding.
 - **RF Coils:** Transmit RF pulses and receive signals from the body.
 - **Computer System:** Processes signals to reconstruct images.
- **Patient Table:** Slides into the bore of the magnet.



IMAGE CONTRAST IN MRI

- **T1-Weighted Images:**
 - Highlights fat and fluid (e.g., used for brain anatomy).
- **T2-Weighted Images:**
 - Highlights water content (e.g., used for detecting edema or inflammation).
- **Proton Density (PD) Images:**
 - Balances T1 and T2 contrast for detailed tissue visualization.
- **Contrast Agents:**
 - Gadolinium-based agents enhance visibility of blood vessels and abnormalities.



APPLICATIONS OF MRI

- **Clinical Applications:**

- Brain and spinal cord imaging (e.g., tumors, multiple sclerosis).
- Musculoskeletal imaging (e.g., torn ligaments, joint abnormalities).
- Cardiovascular imaging (e.g., heart function, blood vessels).
- Abdominal and pelvic imaging (e.g., liver, prostate, uterus).

- **Research Applications:**

- Functional MRI (fMRI) for brain activity mapping.
- Diffusion-weighted imaging (DWI) for stroke assessment.



ADVANTAGES AND LIMITATIONS

- **Advantages:**

- No ionizing radiation.
- Excellent soft tissue contrast.
- Multiplanar imaging (axial, sagittal, coronal views).

- **Limitations:**

- Expensive equipment and maintenance.
- Long scan times (30–60 minutes).
- Contraindications for patients with metallic implants or claustrophobia.



SAFETY CONSIDERATIONS

- **Magnetic Field Hazards:**
 - Ferromagnetic objects can become projectiles.
 - Patients with pacemakers or metal implants may not be eligible.
- **Acoustic Noise:**
 - Loud knocking sounds during scanning; ear protection is required.
- **Contrast Agents:**
 - Rare risk of allergic reactions or nephrogenic systemic fibrosis (NSF) in patients with kidney disease.



RECENT ADVANCEMENTS IN MRI

- **High-Field MRI:**
 - 7 Tesla MRI for ultra-high-resolution imaging.
- **Functional MRI (fMRI):**
 - Maps brain activity by detecting changes in blood flow.
- **Diffusion Tensor Imaging (DTI):**
 - Tracks white matter tracts in the brain.
- **Artificial Intelligence (AI):**
 - AI algorithms for faster image reconstruction and improved diagnostics.



CONCLUSION

- MRI is a powerful, non-invasive imaging tool with diverse clinical and research applications.
- Ongoing advancements continue to improve its capabilities and accessibility.
- **Future Directions:**
 - Portable MRI systems.
 - Enhanced imaging techniques for personalized medicine.

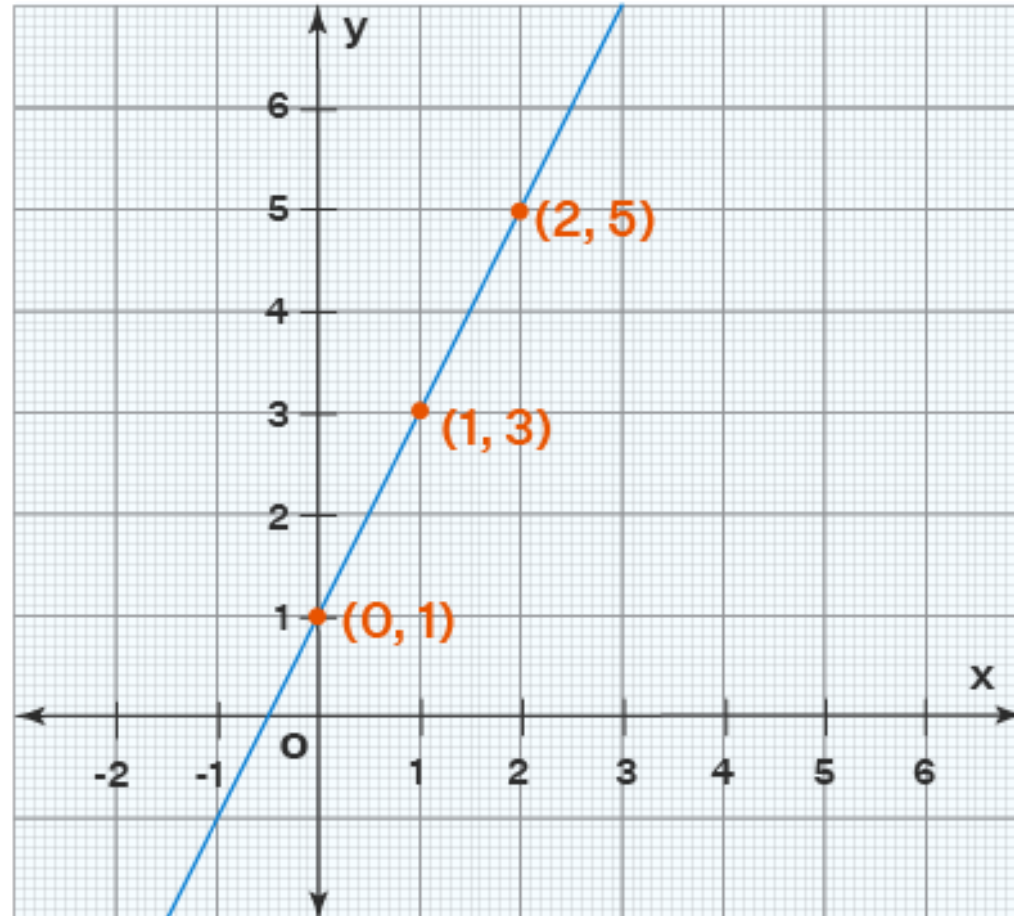


THANKS



X AND Y GRAPH

x	y
2	5
1	3
0	1



HOW DOES AN ULTRASOUND MACHINE WORK?

- The sound waves travel into the body and hit different tissue, fluid or bone
- Some of the sound waves get reflected back to the probe, while some travel on further until they reach another boundary and then get reflected.
- The reflected waves are picked up by the probe and relayed to the machine.



- Using **speed = distance / time** the machine calculates the distance from the probe to the tissue or organ.
- The machine displays the distances and intensities of the echoes on the screen, forming a two dimensional image

$$V = \frac{x}{t}$$



